

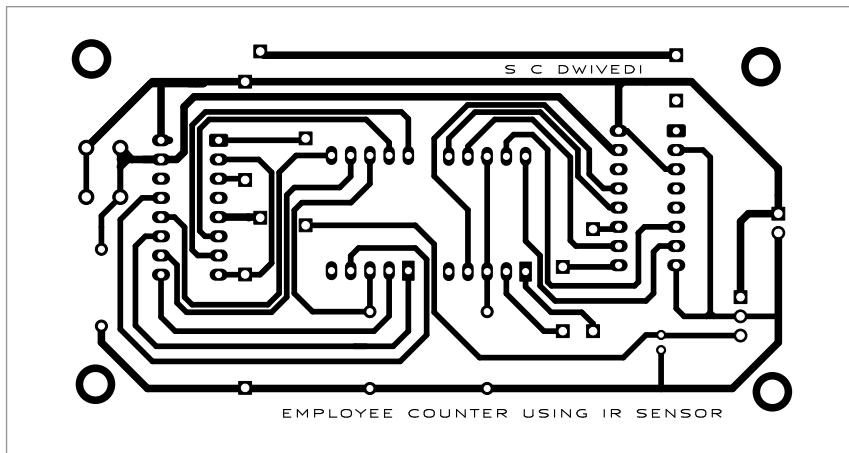


Fig. 3: Actual-size PCB

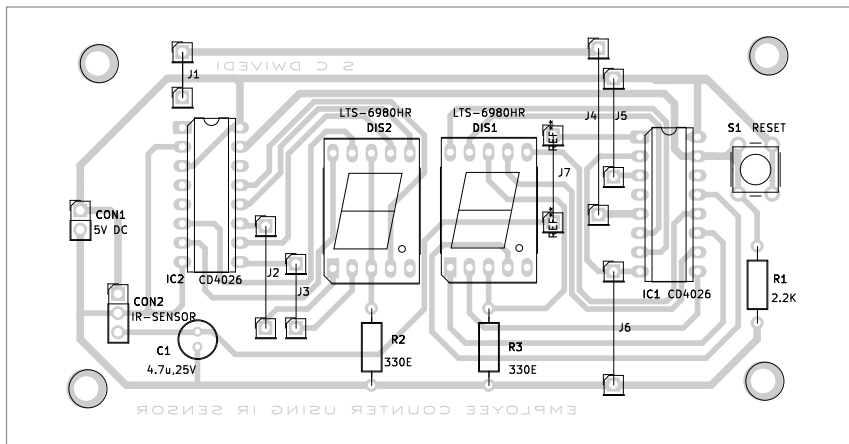


Fig. 4: Components layout

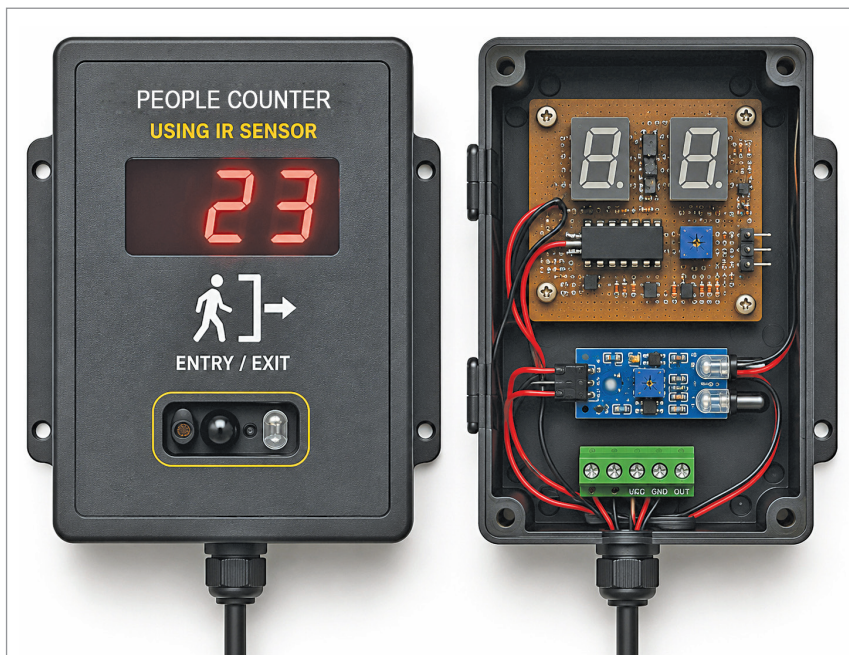


Fig. 5: Suggested wall-mounted enclosure

PARTS LIST

Semiconductors:

IC1, IC2 - CD4026B decade counter/7-segment driver

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 2.2k Ω
R2, R3 - 330 Ω

Capacitors:

C1 - 4.7 μ F, 25V electrolytic

Miscellaneous:

DIS1, DIS2 - 7-segment CC display (LTS6980HR/LTS543)

S1 - Push-to-on switch

CON1, CON2 - 2-pin connector
- IR sensor module
- Designed PCB or general-purpose board
16-pin IC base - 2 (optional)

- Connecting wires
- Enclosure/cabinet
- Screws and spacers
- 5V DC battery or adaptor for power supply

location, such as the entrance or at a security desk, while the circuit should be housed in a suitable protective enclosure. A suggested wall-mounted enclosure with its internal view is shown in Fig. 5 for reference. However, a different enclosure may be selected to suit the installation requirements.

Use a regulated 5V DC power supply, preferably with surge protection or UPS backup, to ensure reliable operation. Before deploying the system, adjust the sensor sensitivity and test the setup by allowing people to pass through the doorway while verifying that the count is displayed correctly. For the best accuracy, ensure that only one person passes through the detection zone at a time. Periodically inspect the sensor alignment and keep the sensor surface clean to maintain consistent and reliable performance. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-create-your-own-employee-counter-using-an-ir-sensor>

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